

Classifying Historical Attitudes

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Abstract

This project studies sentiment classification of historical women's physical education texts in Mount Holyoke's archives, labeling each entry as supportive or unsupportive of physical education. The task is challenging because historical texts often contain OCR errors, inconsistent spelling, and older linguistic patterns that make straightforward text analysis difficult. To address this problem, we cleaned and processed the text data, and then compared three classification approaches: a manually engineered logistic regression model and a TF-IDF-based logistic regression model and a Sentence-BERT-based logistic regression model. The manual model relied on hand-designed features intended to capture relevant language, while the TF-IDF model used word and bigram frequency patterns. The Sentence-BERT model used contextual embeddings to capture richer semantic information. Our results show that Sentence-BERT achieved higher accuracy outperforming the manual and TF-IDF feature-based models. These findings suggest that contextual embeddings are especially useful for noisy historical archival text, while

simpler feature-based approaches still provide meaningful baselines. Overall, this project demonstrates the potential of NLP and machine learning for archival text analysis and suggests future work could improve performance with larger datasets and more advanced models.

1. Introduction

Historical archives contain valuable information about how people wrote about women's physical education and health. In the nineteenth century, debates about women's physical activity and schooling were closely tied to ideas about health, femininity, and social expectations. Some educators and reformers supported women's physical training, while others questioned whether such instructions were appropriate. Since the texts we analyze reflect important historical attitudes, they offer a useful setting for studying how language shows societal sentiments. Manual reading of archival collections is slow and difficult to scale. Researchers can closely interpret a small number of documents, but large collections contain far more material than can reasonably be examined. Digitization makes these archives easier to analyze, but new challenges arise. Historical documents often contain OCR errors, inconsistent spelling, unusual formatting, and older linguistic patterns that make traditional text analysis difficult across multiple documents. These challenges are common in historical NLP research, where spelling variation, document degradation, and OCR noise frequently reduce text quality and complicate computational analysis (Piotrowski, 2012)¹. As a result, historical text classification is a useful test case for natural language processing methods.

In this project, we explore whether NLP can help classify historical women's physical education text from Mount Holyoke's archives as supportive or unsupportive of physical education. Rather than relying on manual close reading, we compare multiple machine learning models: a manually engineered logistic regression model, a TF-IDF based logistic regression model, and Sentence-BERT based approach. By evaluating these models on the same dataset, we study which

¹ Piotrowski, Michael. 2012. "Natural Language Processing for Historical Texts." *Synthesis Lectures on Human Language Technologies* 5(2): 1–157.

methods are most effective for archival text classification.

This task is especially important because it demonstrates how computational methods can support historical research. Instead of reading documents one by one, scholars can use NLP to analyze patterns across large archival collections. At the same time, the task provides a strong test of how well simple machine learning methods can handle real archival text. In the sections that follow, we describe the dataset, explain our methods, present the results, and discuss the implications of our findings for applying NLP to historical archives.

2. Related Work

Prior works in NLP historical text analysis have explored the challenges related with OCR noises, implicit language detection, and sentimental classification. Researchers have showed that computational approaches are helpful in analyzing historical corpora more efficiently than traditional approach of close reading alone.

2.1 Historical Text Digitalization and OCR

One significant challenge in applying NLP methods to historical text analysis is the noise caused by optical character recognition (OCR) digitalization, which is necessary for translating archival text into machine-readable data. OCR noises is primarily caused by low-quality and unrecognizable images. Jentsch et al. summarize several preprocessing strategies to reduce the OCR noises and improve ultimate NLP performance. On the other hand, Lorenzen (2020)² argues that "dirty OCR data" could still produce meaningful computational result because the broader textual pattern remains

detectable. Since our data also consists of OCR-processed text containing potential noise, Lorenzen provides method justification for our project³.

At the same time, we adopted preprocessing strategies suggested by Jentsch et al., such as ensuring the input image quality, to minimize OCR noises.

2.2 Implicit bias in language

The prior NLP works also explored how social ideologies are implicitly encoded in languages. To address this problem, Sap et al.(2020) introduce the Social Bias Framework to analysis the social attitude subtly expressed in text. Sap et al. has proven the limitation of traditional binary classification approach.⁴ Since our project investigate the social attitude about women body training in 19th century, implicit wording frequently occurs. Though our current work focuses on the traditional classification approach, Sap et al. provide the project an important next-step to go.

2.3 Sentimental Classification of Historical Text

There has been researchers conducting similar binary classification analysis on historical text. Al-Laith et al. (2023)⁵ evaluate the performance of multiple classification models on a manually annotated dataset of historical Danish and Norwegian texts labeled with positive or negative categories. Al-Laith et al.'s work is closely related to our project, as both works apply supervised classification methods on manually labeled historical corpora. This prior work suggests that our overall approach is feasible.

Patrick Jentsch and Stephan Porada, "From Text to Data: Digitization, Text Analysis and Corpus Linguistics," in *Digital Methods in the Humanities*, ed. Silke Schwandt (2020), 89–128.

³ Malte Lorenzen, "Testing Hypotheses with Dirty OCR and Web-Based Tools in Periodical Studies," in *Digital Methods in the Humanities*, ed. Silke Schwandt (2020).

⁴ Maarten Sap et al., "Social Bias Frames: Reasoning about Social and Power Implications of Language," in *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (2020), 5477–5490.

⁵ Ali Al-Laith et al., "Sentiment Classification of Historical Danish and Norwegian Literary Texts," in *Proceedings of the 24th Nordic Conference on Computational Linguistics (NoDaLiDa)* (2023), 324–334.

3. Dataset

We used a manually collected and labeled dataset to train and test out classification model. The dataset consists of OCR-processed historical text spanning from 1800 to 1900. Table 1 shows the statistical information about the corpus.

3.1 Corpus Collection

We collect our corpus from writings of authors who are known for their clear positions on women’s physical education. For instance, we select supportive texts on women’s physical education from Mary Lyon and Zilpah Grant, and unsupportive texts from Edward H. Clarke and Pierre de Coubertin. Each data point consists of a short text segment, all of which are manually selected to capture clear stance related to women physical education. Additionally, we manually reviewed OCR-processed text to correct recognition errors caused by low-quality archival scans.

3.2 Annotation and Label

Texts with a clear ideology stance are manually identified and labeled. Supportive texts generally frame women’s exercise as beneficial for health, body, and moral development. Unsupportive texts portray women’s physical education to be dangerous, harmful, and inappropriate. Only the selected text segment is considered during labeling. Contextualization on what might go on before and after the segment is ruled out.

Label	Count	Percentage
Supportive	50	55%
Unsupportive	41	45%
PE-related texts	91	78%
Non-PE texts	26	22%

Table 1: Dataset Statistics overview

In addition to the stance label, the dataset includes a binary relevance label (`is_pe`) indicating whether the text is related to women’s exercise or physical training. This additional annotation layer is introduced to support future development of this project, serving as an

automatic filter to identify physical education related text before manual annotation.

3.4 Challenge and Limitation

Besides the general implicit nature of 19th century texts, women’s education and physical bodies were sensitive and controversial topics in this time period, causing most related texts to express attitudes in very subtle, indirect ways. Authors who support women’s exercise may also acknowledge physical limitations of women’s bodies, while those opposed to physical education may still emphasize women’s intellectual capabilities. Therefore, although an author’s stance often signals the general attitude of a text, careful manual annotation is required for each data point to ensure that the model does not misclassify nuanced cases.

4. Methods and Models

4.1 Data Preprocessing

Before training the model, we preprocessed the archival data to ensure consistency and usability for classification. We removed rows that lacked text or had missing labels. We did this to ensure that each example contained both a document and a category. We then cleaned the labels by converting them to lowercase, removing unnecessary punctuation, and stripping extra whitespace. After cleaning, we mapped the labels to a binary format, with supportive text assigned to one class and unsupportive text to the other. We also split the data into training and test sets so that the model would use at least 80% of the data to train and the remaining 20% to evaluate performance on unseen data. The text was also converted to lowercase before feature extraction to reduce variation caused by capitalization. We tried not to remove the language patterns aggressively, in case they still contained useful information for classification. Instead, we relied on feature-extraction methods to obtain relevant information from the text. Overall, preprocessing ensured that the input data was clean, consistent, and suitable for training and evaluating both models.

4.2 Logistic regression model

All three feature representations were evaluated using the same logistic regression classifier. Logistic regression estimates the probability that a text belongs to the supportive class by taking a

weighted combination of the input features and passing it through a sigmoid function. We trained the classifier with gradient descent, starting from weights initialized to zero and updating the weights and bias iteratively to reduce binary cross-entropy loss. We also standardized the feature values using the training set mean and standard deviation, and we stopped fairly early if the change in loss became sufficiently small to prevent unnecessary overfitting. A threshold of 0.5 was used to convert predicted probabilities into binary class labels. This made it easy to apply the same classifier to different feature representations, including manual features, TF-IDF vectors, and Sentence-BERT embeddings.

4.3 Manual Feature model

We implemented a manual feature-based model to capture simple lexical and stylistic features that might show whether a text is supportive or unsupportive. For each document, we extracted a small set of manual features: the number of positive cue words, the number of negative cue words, the presence of negation, and the total number of words in the text. The positive list included terms like exercise, gym, athletic, health, strong, fitness and benefit, while the negative list had words like weak, fragile, unhealthy, sick, injury, pain. We handled simple negation, marking occurrences of “not” and “no” before feature extraction. We did this because negation can heavily influence the meaning of a statement, making it positive or negative. These features were then used as an input to a logistic regression classifier trained with gradient descent. Before training, the features were standardized using the mean and standard deviation of the training set, and the model was trained for up to 1000 iterations. Since the model rely on simple word counts and surface-level pattern, its predictions are directly interpretable.

4.4 TF-IDF model

We used TF-IDF feature representation to extract informative terms from text. This method does not rely on a predefined word list but converts each document into a weighted vector based on term frequency and relative importance of terms across the entire corpus. This weighting approach is based on traditional TF-IDF methods commonly used in information retrieval and text classification (Salton and Buckley, 1988)⁶. We

used a TF-IDF vectorizer with English stop words removed, unigrams and bigrams included, and a maximum vocabulary size of 1000 features. Including the bigrams allowed the model to capture short contextual phrases, while limiting the vocabulary kept the feature space more manageable and reduce noise from less informative terms. The TF-IDF vectors were then used as input to the same logistic regression classifier trained with gradient descent. We included this model because it could learn from a wider range of words and phrases without requiring hand-engineered feature lists.

4.5 PE Filter model

In addition to the sentimental classification task, which is the project’s main focus, we also implemented a separate binary classification model to identify whether a given text is related to Physical Education(PE). The model classifies each text segment into either PE-related (`is_pe=1`) or non-PE (`is_pe=0`).

The PE model is implemented using a similar approach to the manual sentiment model. Specifically, we created two lists of manually extracted features for PE-related keywords (Ex: ‘gym’, ‘exercise’, ‘outdoor’), and non-PE keywords (Ex: ‘bible’, ‘history’, ‘composition’). A logistic regression classifier is trained on these features and their related discourse in physical education, resulting in a filter model serving as an automatic preprocessing layer for the sentimental classification model.

The current PE-labeled dataset is imbalanced, with a considerably larger proportion of PE-related data than non-PE data. This is also reflected in the model’s evaluation. Although the PE classifier model has relatively high accuracy of 0.79 on the test set, the confusion matrix indicates that the model has a strong bias toward the positive category, and isn’t able to classify any non-PE data correctly.

Despite the imbalance, PE model functions as expected, suggesting that performance issue is mainly due to the dataset rather than model itself. In future practice, the PE filter model could potentially reduce the burden of manually identifying PE-related data from large-scaled corpora and enable more efficient sentimental analysis.

4.6 Sentence-BERT model

⁶ Salton, Gerard, and Christopher Buckley. 1988. “Term-Weighting Approaches in Automatic Text Retrieval.” *Information Processing & Management* 24(5): 513–523.

This model converts each text into a dense contextual embedding, meaning it represents the overall meaning of the sentence and not just individual word counts. We used the all-mpnet-base-v2 Sentence-BERT model to convert each text entry into a 768-dimensional numerical representation. These embeddings were then used as input into the same logistic regression classifier trained with gradient descent. We included the Sentence-Bert model because it can recognize relationships between words and meanings that may not be captured through keyword counts or TF-IDF features alone. This was especially useful for historical texts where the sentiments may not be directly expressed. That being said, the model was pre-trained on modern English corpora, OCR errors and linguistic patterns in the archival text may still affect effectiveness.

Manual features achieved the strongest balance across metrics with accuracy of 74%. The TF-IDF model achieved lower overall performance with an accuracy of 58% and lower recall for the supportive class. The Sentence-BERT model achieved the highest overall accuracy at 84%, outperforming both the manual feature and TF-IDF approaches.

4.7 Evaluation

To evaluate model performance, we split the dataset into training and test sets, using 80% of the data for training and 20% for testing. The test set was held out during training so that each model could be evaluated on unseen data. We evaluated performance using accuracy, precision, recall, F1 score and confusion matrices. Accuracy measure the overall percentage of correctly classified texts. Precision measured how often the model's predictions for a class were correct. Recall measured how many examples of that class the model correctly identified. F1-score combined precision and recall into a single metric. We also used confusion matrices to examine patterns of correct and incorrect predictions for each class. These metrics provided a fuller view of model performance.

5. Results

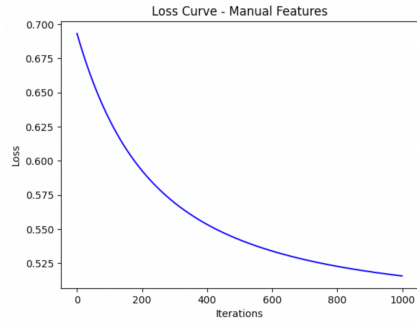
5.1 Overall Model Performance

Models were evaluated on the held-out test set using accuracy, precision, recall, F1-score, confusion matrices. Table 1 summarizes the performance of the manual feature, TF-IDF, and Sentence-BERT models.

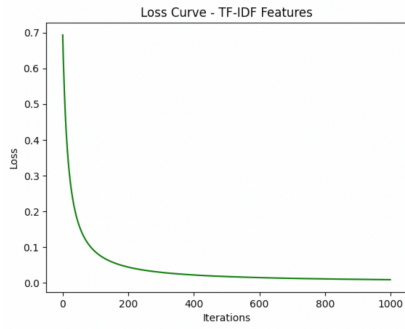
Table 2: Sentiment Classification Results

Model	Accuracy	Macro F1	Precision (0/1)	Recall (0/1)
Manual Features	0.74	0.74	0.70/0.78	0.78/0.70
TF-IDF	0.58	0.57	0.54/0.67	0.78/0.40
Sentence-BERT	0.84	N/A	N/A	N/A

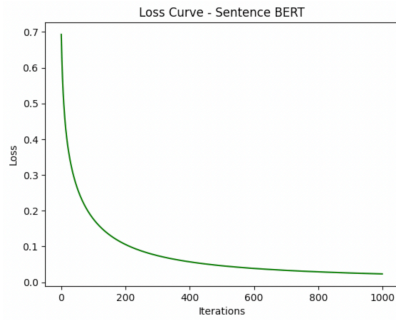
5.2 Training Loss curves



a. manual feature model



b. TF-IDF Model



c. Sentence-BERT Model

Figure 5.2 a,b,c: Training loss curves for the manual feature, TF-IDF and Sentence-Bert logistic regression models. All models show decreasing loss during training, indicating successful optimization. The TF-IDF and the Sentence-BERT models converged more rapidly and reached lower final loss values than the manual feature model.

5.3 PE Filter Model Results

Table 3: PE Filter Performance

Metric	Value	Details
Accuracy	0.79	19/24 correct
Macro F1	0.44	Penalized by class imbalance
PE Recall	1.00	Perfect PE detection
Non-PE Recall	0.00	No non-PE texts identified

In addition the main sentiment classification task, we evaluate a separate PE filter model to classify whether a document was related to physical education. The PE filter model was evaluated on a held-out test set containing 24 examples, including 19 PE-related texts and 5 non-PE texts

Table 4: Confusion matrix for the PE filter model

Actual / Predicted	Non-PE	PE
Non-PE	0	5
PE	0	19

The PE matrix shows that the PE filter model correctly identified all PE related texts but failed to correctly classify any non-PE examples. All five non PE examples were incorrectly predicted as PE-related, while all 19 PE documents were classified correctly. This resulted in perfect recall for the PE class but poor performance of the non-PE class

6. Discussion and Improvement

Based on the evaluation results, we conducted error analysis and explored corresponding improvement on TF-IDF and manual feature based models.

6.1 Feature Simplification.

Another attempt we made to improve the performance of the manual feature model is to simplify the features that contributed little to model classification. The initial manual feature model included features such as *pronoun_count* and *has_modal*. Based on the feature weight coefficients learned by the logistic regression classifier (shown in Table 2), sentiment-related

Model Stage	Accuracy
TF-IDF	0.58
Manual Features (Baseline)	0.63
+Feature Simplification	0.68
+Negation Handling	0.74
Sentence-BERT	0.84

Table 5: Model accuracy across different stages, including the initial TF-IDF model, the initial manual feature-based model, the manual feature model with feature simplification, and the model with negation handling. Ultimately, the S-BERT model outperforms all other approaches.⁷ Conclusion

features such as positive word counts (0.978) and negative word counts (-0.542) contributed strongly to model prediction. In contrast, pronoun counts (0.032) and modal verbs (-0.100) display relatively small coefficient values, indicating their limited contribution.

Based on the evaluation, we removed the *pronoun_count* and *modal_verb* features from the manual feature list. After feature simplification, the manual feature model accuracy improved from 0.63 to 0.68. This result suggests that reducing weak or noisy features can improve the overall performance of classification models.

6.2 Negation Handling.

Prior works in sentiment analysis has shown that negation handling is a simple but effective strategy for improving sentiment classification performance by capturing the polarity reversing in texts. Specifically, Kiritchenko, et al.(2014) demonstrated that correctly capturing sentiment features in a negated context could substantially improve model classification accuracy⁷.

Inspired by Kiritchenko, et al., we attempted an additional negation preprocessing to convert phrases such as ‘not good’ into ‘NOT_good’. This allows the model to preserve the negated semantics, rather than just treating negation word counts as an independent variable separated from their surrounding context.

After implementing negation preprocessing, the manual feature model indicated an evident accuracy improvement from 0.68 to 0.74. This suggests that explicit negation handling is

effective in reducing sentiment misclassification in historical text classification.

6.3 Increasing ngrams and TF-IDF Tuning.

In addition to feature simplification and negation handling, we also conducted some experiments aimed at improving the TF-IDF model through parameter tuning and increasing contextual representation. We tried to increase the n-gram range from unigrams and bigrams to trigrams, and adjust a few vectorization parameters, including increasing the maximum feature size (`max_features=5000`) and removing infrequent terms (`min_df=2`). Although these adjustments were trying to improve the model performance, the classification accuracy remained almost unchanged. This suggests that unigram and bigram have captured most of the possible contextual information in the dataset, and the additionally higher n-grams barely contribute. The failing attempts to improve TF-IDF motivated our exploration of the Sentence-BERT contextual embeddings approach in the project.

6.4 Sentence-BERT model

To address the sparse semantic representations problem, we explored the Sentence-BERT (SBERT) to derive semantically meaningful vectors, where similar texts are placed closely in the vector space. Compared to traditional BERT, SBERT uses siamese and triplet network structures that enable cheaper and more efficient contextual embedding generation that are suitable for historical text analysis.⁸

SBERT computes the contextual relationships between words rather than direct word counts. We used the `all-mpnet-base-v2` model to convert each text segment into a 768-dimensional dense vector representation. The embeddings created by the model are then used as input to the same logistic regression classifier that we have been using for the previous classification models, which enable a controlled environment for us to compare - performance with the other models.

Compared to TF-IDF and manual feature models, Sentence-BERT is able to capture similar semantics information and contextual dependencies in the text, which makes it very

⁷ Svetlana Kiritchenko, Xiaodan Zhu, and Saif M. Mohammad, “Sentiment Analysis of Short Informal Texts,” *Journal of Artificial Intelligence Research* 50 (2014): 1–15.

helpful for historical text analysis, in which stance and idea are implicitly hidden within the texts.

7. Conclusion

In this project, we investigated the application of sentiment classification on historical texts from Mount Holyoke's archives. By constructing a binary classification between supportive and unsupportive stances towards women's physical education, we want to know whether NLP methods can reliably identify implicit sentiments in 19th century corpora. We manually collected and labeled a dataset while navigating challenges such as OCR noises and ambiguous expression of stance. To address these challenges and to investigate our question, we evaluated three approaches: TF-IDF logistic regression, manually designed feature logistic regression, and Sentence-BERT contextual embedding. We further conducted several iterative improvements especially on the manual feature model, including feature simplification and negation preprocessing. These modifications resulted in positive improvement in model performance.

Our results show that TF-IDF provides a baseline standard that is restricted by the sparse lexical representations that couldn't capture deeper meaning in the text. The manual feature model's improvements through feature adjustment and negation addressing also demonstrates that a carefully-designed feature set can still be effective in small dataset or corpora of specific thematic context. Sentence-BERT model achieves the best overall performance, proving the strength of contextual embedding in capturing the implicit ideologies in historical texts.

In summary, we have gained both in-depth knowledge and practical experience in applying NLP technique on archival materials. The project displays the great potential of NLP computational approaches to complement the traditional close-reading research approach in humanity and historical research. Future work of this project may explore the fine-tuning of S-BERT on historical corpora, developing more efficient strategies supported by the PE filtering model to assist preprocessing and annotation. We are looking forward to the day when dataset and models could uncover patterns in historical discourse that humans could not discover.

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